

RV32I Base Integer ISA — Quick Reference

Opcode / funct3 / funct7 tables + immediate encodings, for control unit design

Base Instruction Formats

Format	31:25	24:20	19:15	14:12	11:7	6:0
R	funct7	rs2	rs1	funct3	rd	opcode
I	imm[11:0]		rs1	funct3	rd	opcode
S	imm[11:5]	rs2	rs1	funct3	imm[4:0]	opcode
B	imm[12,10:5]	rs2	rs1	funct3	imm[4:1,11]	opcode
U	imm[31:12]				rd	opcode
J	imm[20,10:1,11,19:12]				rd	opcode

Immediate Bit Construction (sign-extend always uses inst[31])

Type	Bit layout (MSB → LSB)
I	{inst[31] x20, inst[31:20]}
S	{inst[31] x20, inst[31:25], inst[11:7]}
B	{inst[31] x19, inst[31], inst[7], inst[30:25], inst[11:8], 1'b0}
U	{inst[31:12], 12'b0}
J	{inst[31] x11, inst[31], inst[19:12], inst[20], inst[30:21], 1'b0}

R-Type — opcode 0110011 (OP)

funct3	funct7	Instruction	Operation
000	0000000	ADD	rd = rs1 + rs2
000	0100000	SUB	rd = rs1 - rs2
001	0000000	SLL	rd = rs1 << rs2[4:0]
010	0000000	SLT	rd = (rs1 < rs2) signed
011	0000000	SLTU	rd = (rs1 < rs2) unsigned
100	0000000	XOR	rd = rs1 ^ rs2
101	0000000	SRL	rd = rs1 >> rs2[4:0] (logical)
101	0100000	SRA	rd = rs1 >>> rs2[4:0] (arithmetic)
110	0000000	OR	rd = rs1 rs2
111	0000000	AND	rd = rs1 & rs2

I-Type — opcode 0010011 (OP-IMM)

funct3	funct7 (shifts only)	Instruction	Operation
000	—	ADDI	rd = rs1 + imm
010	—	SLTI	rd = (rs1 < imm) signed
011	—	SLTIU	rd = (rs1 < imm) unsigned
100	—	XORI	rd = rs1 ^ imm
110	—	ORI	rd = rs1 imm

funct3	funct7 (shifts only)	Instruction	Operation
111	—	ANDI	rd = rs1 & imm
001	0000000	SLLI	rd = rs1 << shamt
101	0000000	SRLI	rd = rs1 >> shamt (logical)
101	0100000	SRAI	rd = rs1 >>> shamt (arithmetic)

Loads — opcode 0000011 (LOAD)

funct3	Instruction	Operation
000	LB	rd = sign_ext(mem[rs1+imm][7:0])
001	LH	rd = sign_ext(mem[rs1+imm][15:0])
010	LW	rd = mem[rs1+imm][31:0]
100	LBU	rd = zero_ext(mem[rs1+imm][7:0])
101	LHU	rd = zero_ext(mem[rs1+imm][15:0])

Stores — opcode 0100011 (STORE)

funct3	Instruction	Operation
000	SB	mem[rs1+imm][7:0] = rs2[7:0]
001	SH	mem[rs1+imm][15:0] = rs2[15:0]
010	SW	mem[rs1+imm][31:0] = rs2[31:0]

Branches — opcode 1100011 (BRANCH)

funct3	Instruction	Condition (branch taken if...)
000	BEQ	rs1 == rs2
001	BNE	rs1 != rs2
100	BLT	rs1 < rs2 (signed)
101	BGE	rs1 >= rs2 (signed)
110	BLTU	rs1 < rs2 (unsigned)
111	BGEU	rs1 >= rs2 (unsigned)

Jumps and Upper Immediate

Instruction	Opcode	funct3	Operation
JAL	1101111	—	rd = PC+4; PC = PC + imm(J)
JALR	1100111	000	rd = PC+4; PC = (rs1 + imm(I)) & ~1
LUI	0110111	—	rd = imm(U)
AUIPC	0010111	—	rd = PC + imm(U)

Reference: RISC-V Unprivileged ISA Spec, RV32I Base Integer Instruction Set